Day_08_Mini_Cybersecurity_Challenge_#1

TASK 1 — NETWORK IDENTITY (20 POINTS)

Run ipconfig /all and record: IPv4 Address · Subnet Mask · Default Gateway · DNS Server · MAC / Physical Address · DHCP Enabled (Yes/No).

IPv4 Address: 10.101.139.72

Subnet Mask: 255.255.224.0

Default Gateway: 10.101.128.1

DNS Server: 10.101.128.1

MAC/ Physical Address: 24-B2-B9-9F-EC-75

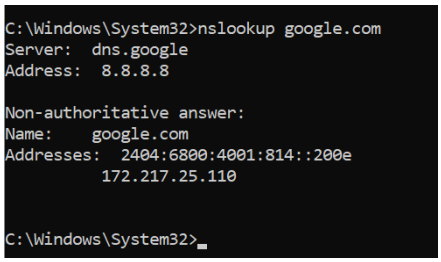
DHCP Enabled: YES

TASK 2 — NETWORK LAYER MATCHING (15 POINTS) - COMPLETED

TASK 3 — IP ADDRESS CLASSIFICATION (15 POINTS) – COMPLETED

TASK 4 — DNS INVESTIGATION (15 POINTS)

Run: nslookup google.com



1) Which DNS server responded?
   DNS Server: google.com
   IP Address: 8.8.8.8
2) Which IP address was returned?
   IPv4 Address: 172.217.25.110
   IPV6 Address: 2404:6800:4001:814::200e
3) What is the purpose of DNS?
   DNS stands for Domain Name System
   It translates domain name such as google.com into IP addresses, so it makes it easy for computers to locate and communicate with servers.

Day_08_Mini_Cybersecurity_Challenge_#1

TASK 5 — NETWORK PATH INVESTIGATION (15 POINTS)

Run: tracert google.com

```
C:\Windows\System32>tracert google.com

Tracing route to google.com [172.217.25.110]
over a maximum of 30 hops:

  1    29 ms     6 ms     2 ms  10.101.128.1
  2    28 ms    32 ms     7 ms  192.168.200.1
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22     *        *        *     Request timed out.
 23     *        *        *     Request timed out.
 24     *        *        *     Request timed out.
 25     *        *        *     Request timed out.
 26     *        *        *     Request timed out.
 27     *        *        *     Request timed out.
 28     *        *        *     Request timed out.
 29     *        *        *     Request timed out.
 30     *        *        *     Request timed out.

Trace complete.

C:\Windows\System32>
```

1) How many visible hops appeared: 2 visible hops responded

2) What is the first hop: 10.101.128.1

3) What device commonly represents the first hop: Default Gateway, usually a router.

4) What is traceroute/tracert used for?

It is used to identify route packets to reach destination and helps with troubleshooting network connectivity issues.

TASK 6 — FINAL FLAG CHALLENGE (20 POINTS)

Scenario — a system has this configuration:

IPv4 Address: 192.168.50.25

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.50.1

MAC: AA-BB-CC-DD-EE-FF

Network Address: 192.168.50.0

Broadcast Address: 192.168.50.255

Day_08_Mini_Cybersecurity_Challenge_#1

First Usable Host: 192.168.50.1

Last Usable Host: 192.168.50.254

Answer: JVL{192_168_50_0}

